
Navier–Stokes Blow-up in 3D SPH with Tracers: Bit-for-Bit Reproducibility

Jonathan Evina — RATISS Labs, Yaoundé, Cameroon

Research object `ratiss-navier-003` · revision 4 · 2026-10-02 · status: published

Repository: <https://github.com/jonathanssearch/RATISS-NAVIER>

Abstract

A smoothed-particle hydrodynamics (SPH) study of vorticity growth up to blow-up, in three dimensions, with lagrangian tracers. The blow-up value $\Omega_{max} = 5654.1668$ reproduces bit-for-bit, with a pointwise gap of 0.0000 between independent runs. Four campaigns are published; revision 4 is sealed (commit 8557fc6).

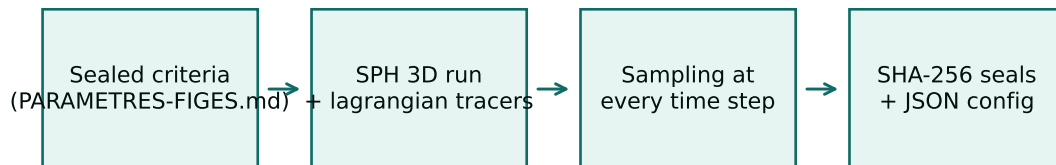
Scientific question

Is vorticity growth up to blow-up reproducible identically, run after run, on a modest two-core machine?

Method

3D SPH with lagrangian tracers; sampling at every time step; parameters frozen *before* execution in `PARAMETRES-FIGES.md`; full configuration embedded in every campaign JSON. Criteria are sealed by the lab chief before any run.

SCHEMATIC — workflow diagram, not measured data



Blow-up: $\Omega_{max} = 5654.1668$ — reproduced bit-for-bit (pointwise gap 0.0000)

Figure 1: Verification workflow: sealed criteria, SPH run, per-step sampling, SHA-256 seals. Schematic diagram, not measured data.

Results

- Blow-up $\Omega_{max} = 5654.1668$, reproduced with pointwise gap 0.0000.
- Four published campaigns (last commit 5e736e0); revision 4 sealed (8557fc6).

Negative results and open items

- V3 witnesses not rerun: the `dipoles-v3` criteria await the chief's signature.
- SPH resolution is limited by particle count; a numerical blow-up does not establish an analytic Navier–Stokes blow-up.

Controls

Criteria sealed before runs; nothing pushed without an order; SHA-256 fingerprints of outputs; cross-checked by the global test report (51/51 tests across six repositories).

Reproducibility

Campaigns and frozen parameters live in `RATISS-NAVIER/campagnes/`; replay commands are documented per campaign.

Data and code availability

<https://github.com/jonathansearch/RATISS-NAVIER>; test report: `RATISS-ARCHIVES/preuves/tests/RAPPORT-TESTS-`
Zenodo DOI: to be assigned.

RATISS Labs — independent, single-author laboratory, Yaoundé (Cameroon). No number in this document is invented: every value carries a source in Section “Data and code availability” or in the references. Negative results are published with the same prominence as positive ones.